ARENDA RESEARCH INITIATIVE

WHITEBOX v2 — Agent Team Handbook

The Internationals Sprint: every experiment, operationally

Arjun Paradkar · September 2026 · For Claude Code agents executing the program

1 Read This First — How Agents Operate

1.1 The program in one paragraph

Nobody can currently tell which explanations of an AI are real. Networks have symmetry groups — neuron permutations and weight rescalings that change nothing about what the network computes — and no interpretability method has ever been audited against them. This is the mechanistic extension of a 2017 axiom the field already accepts for input attributions (Sundararajan et al., Implementation Invariance) but has never applied to its own internals. We build the audit, run it, ship it, and use the certified measurements to attack the black-box problems that survived scrutiny. The mathematical foundation is topological in the deep sense: invariants under a transformation group, connected components of a quotient space, cohomological obstructions to gluing. Persistent homology is one instrument among several.

1.2 Non-negotiable rules

1. **SPEC before compute.** Every experiment directory gets `SPEC.md` with pre-registered hypotheses and numeric pass/fail thresholds *before* any run. Show it to the principal at the pause point. Never edit a SPEC after results exist; append `AMENDMENTS.md` instead.
2. **Kill before you believe.** Every positive result runs its listed kill tests (nulls, controls). Report the kill attempts in `FINDINGS.md` whether they succeeded or failed.
3. **Baseline or no claim.** Any "better" or "first" requires the named existing method run on identical data. No baseline row in the results table = the claim is deleted.
4. **Negatives are deliverables.** A null result gets the same figures, tables, and write-up as a positive.
5. **Novelty status is data.** Each experiment carries a status (OPEN / FLAGGED / TAKEN). TAKEN experiments run as anchors or replications and make no novelty claim. FLAGGED experiments require a targeted literature search logged in `NOVELTY.md` before any write-up.
6. **Lab boundaries.** Nothing touching Fiete-lab data runs in this sprint. Public data only.
7. **Result beats framing.** If data contradicts the story, rewrite the story.
8. **Pause points.** Stop and report at the end of each block marked PAUSE. Do not proceed past a pause without principal review.

1.3 Repository layout

```
whitebox-v2/
  whiteboxtopo/ # the library (import name; v1 'whitebox' collides on PyPI)
    core/ # PH backend (giotto-ph default), PCA, bootstrap, FilmRecorder (from v1)
    symmetry/ # gauge group: permutations, scalings, hidden-symmetry detection
    invariants/ # check_invariance(), candidate invariant generators
    nulls/ # shuffle, permutation, Marchenko-Pastur, planted-structure
    graphs/ # attribution graphs, modularity, treewidth heuristic
    discovery/ # fixed circuit-discovery pipeline
    signals/ # memorization_signal (v1), legibility_score, residual_structure
    topology/ # zigzag, multiparameter, Mapper, graph-PH (Z-series)
  experiments/
    F_gauge/ M_memorization/ C_concepts/ D_attribution/ B_bias/
    K_darkmatter/ V_legibility/ Z_topology/
```

```
    each: SPEC.md run_*.py results/{csv,fig} FINDINGS.md NOVELTY.md
benchmarks/ # batteries, calibration, validation
CLAIMS.md AUDIT.md VALIDATION.md NOVELTY_BOARD.md
```

1.4 Standard file outputs

Every experiment writes: `results/*.csv` (all numbers), `results/fig/*.png` (300 dpi, colorblind-safe, one figure per pre-registered hypothesis), `FINDINGS.md` (structure: Question / Setup / Pre-registered thresholds / Result table / Kill-test outcomes / Verdict / What this licenses us to claim / What it does not), and a one-paragraph plain-English summary at the top of `FINDINGS.md`.

1.5 Compute legend

[LAPTOP] CPU, run immediately. **[5070]** local RTX 5070 (8 GB). **[HPCC]** Arkansas cluster — these tests convert toy-scale truth into frontier-scale truth. HPCC jobs should be written to run unattended with checkpointing and a resume flag.

2 Shared Infrastructure — build these first

S1 — Symmetry module [LAPTOP]

Purpose. One implementation of the gauge group used by every experiment.
Build. `symmetry.permutations(model)`: within-layer hidden-unit permutations (MLP) and head permutations plus within-head basis rotations (attention). `symmetry.scalings(model)`: positive rescaling through every homogeneous nonlinearity with compensating downstream rescale. `symmetry.apply(model, g)`: returns the transformed model. `symmetry.hidden(model, inputs)`: Jacobian of (parameters → outputs on all inputs); returns corank and the null-space directions.
Tests. For 1000 random g on 10 random models, max output deviation $< 10^{-6}$. Hidden-symmetry detector recovers a planted extra symmetry on a hand-built network.
Output. `whiteboxtopo/symmetry/`, unit tests green.

S2 — `check_invariance` [LAPTOP]

Purpose. The auditor. Given any function `metric_fn(model) -> array`, sample n renamings, return per-component coefficient of variation, a verdict (`invariant` if CV < 0.01, `covariant` if CV scales predictably with the scaling factor, `artifact` otherwise), and the full sample.
Build. Separate permutation-only and scaling-only samples so covariance under scaling can be diagnosed. Include `align=True` option: Procrustes/permutation alignment before comparing, for direction-valued metrics.
Tests. Weight-product anchor → `invariant`. Per-neuron activation → `artifact`. Singular values → `invariant` (perm) / `covariant` (scale).
Output. `whiteboxtopo/invariants/check.py`.

S3 — Nulls library [LAPTOP]

Build. `nulls.feature_shuffle(X)` (destroys joint structure, keeps marginals — the null that killed v1's TC). `nulls.label_permute`. `nulls.marchenko_pastur(spectrum, aspect, var)` returning bulk edges and Tracy–Widom threshold. `nulls.planted(kind, size)` generating data with known structure for calibration.
Tests. MP on Gaussian noise → 0 outliers at $p < 0.01$ over 100 trials.

S4 — Attribution graphs + discovery pipeline [5070]

Build. `graphs.attribution_graph(model, data, threshold)` via attribution patching. `graphs.modularity(G)` (Leiden), `graphs.treewidth_ub(G)` (min-fill), `graphs.sparsity(G)`. `discovery.find_circuit(model, task, faithfulness=0.9)`: attribution patching + greedy prune, returns circuit, size, faithfulness, wall time; deterministic given seed.
Rule. This pipeline is frozen once E4 starts. No per-model tuning.

S5 — Topology backend [LAPTOP]

Build. Wrap v1's giotto-ph pipeline (95th-percentile threshold is lossless per v1 Phase 1; landmark sparsification ships but is unused by signals). Add: `topology.zigzag(clouds_over_time)` (via Dionysus or GUDHI zigzag), `topology.multiparam(cloud, params)` (via RIVET or a 2-parameter rank-invariant approximation), `topology.mapper(cloud, lens, cover)` (KeplerMapper), `topology.graph_ph(G)` (clique-complex persistence on attribution graphs).
Tests. Zigzag on a synthetic cloud with a planted appearing/disappearing loop recovers the correct birth/death interval.

PAUSE 0 — report S1–S5 test results before any experiment runs.

3 Block F — The Spine: Gauge Orbits and the Invariance Audit

Problem attacked: G1 (which explanations are real). **Topological statement:** valid explanations modulo gauge form a quotient space; we compute its connected components (H_0). **NOVELTY: OPEN** (bridge between parameter-symmetry theory and Méloux non-identifiability verified unclaimed; audit framed as the mechanistic extension of Implementation Invariance, which is cited).

F1 — Specimens [LAPTOP]

Procedure. MLPs, ReLU, depth $\in \{2, 3\}$ hidden layers, width $\in \{4, 6, 8, 10\}$. Targets: XOR3, MAJ3, PARITY3, then all 16 non-degenerate 4-input functions of a fixed list. Adam, full-batch on the truth table, train until zero error or 20k steps. $\geq$20 seeds per (function, depth, width).
Output. `results/specimens.csv` (function, depth, width, seed, converged, path). Drop non-converged; log them.

F2 — Enumerate explanations [LAPTOP]

Procedure. For each specimen: (a) all hidden-unit subsets; keep those reproducing the full truth table under mean-ablation of the complement (`circuits_mean`) and separately under zero-ablation (`circuits_zero`); (b) for each circuit, enumerate gate assignments: unit u implements gate g iff $\mathbb{1}[a_u(x) > 0]$ matches g's truth table over all inputs (allow negation); an interpretation is valid if the assigned gates compose to the target.
Output. `results/explanations.csv`: specimen, criterion, circuit_id, interpretation_id. Report $|\text{mean} \cap \text{zero}|/|\text{mean} \cup \text{zero}|$ per specimen.
Kill. Hand-built XOR network with exactly one known circuit: enumeration must return it and only it under both criteria.

F3 — Gauge group per specimen [LAPTOP]

Procedure. S1 permutations + scalings; S1 hidden-symmetry corank. Record `known_dim` vs `fiber_dim`. If `fiber_dim` > `known_dim`, extract null-space directions, attempt to identify them (report as "hidden symmetry, unclassified" if not), and mark specimen `G_incomplete`.
Output. `results/symmetry.csv`.

F4 — Orbit partition (H_0 of the quotient) [LAPTOP]

Definition (pre-register exactly this). Explanations E, E' are gauge-equivalent iff some $g \in G$ maps the unit-set and weights of E onto those of E' (permutation: relabeled unit set matches; scaling: same support, rescaled weights match within tolerance 10^{-6}; interpretations must map gate-to-gate). Compute the partition by union-find over pairs.
Output. `results/orbits.csv`: specimen, criterion, N explanations, K orbits, K/N.
Kill. Planted-mechanism networks with exactly 1, 2, 3 genuinely distinct algorithms (e.g., XOR via two different gate decompositions, wired in parallel and selected by a fixed input) must yield $K = 1, 2, 3$.

F5 — Hypotheses [LAPTOP]

Pre-registered. H-F5a: median $K/N \leq 0.5$ across specimens. H-F5b: regressing $\log K$ and $\log N$ on width, slope($\log N$) $-$ slope($\log K$) > 0 with bootstrap CI excluding 0.
Figures. `fig/F5_KN_vs_width.png` (log-log, per function), `fig/F5_KN_hist.png`.
Kill. Both ablation criteria must give the same qualitative verdict or the disagreement is reported as the headline.

F6 — Orbit autopsy [LAPTOP]

For every specimen with $K > 1$: extract one representative per orbit, render its gate interpretation, and name the algorithm. Deliver a table of distinct algorithms per function. If a "distinct" orbit pair is related by a symmetry not in G, add that symmetry to S1 and rerun F3–F5 (log as amendment).

F7 — Transformer extension [5070]

1–2 layer attention models (d=32, 4 heads) on the same Boolean tasks encoded as token sequences. Extend G with head permutations and within-head rotations (Q, K by R, $R^{-\top}$; V, O likewise). Circuits are head-and-neuron subsets. Repeat F2–F5.
Output. `results/orbits_transformer.csv`.

PAUSE 1 — report F1–F7. This decides the paper's headline.

F8 — Renaming samples [LAPTOP]

For each F1 specimen, 100 permutation-only, 100 scaling-only (log-uniform $\alpha \in [0.1, 10]$), 100 combined renamings. Verify function identity on every sample.

F9 — Anchor invariants [LAPTOP]

Input→output weight product; expected CV $< 10^{-6}$. This is the pipeline sanity gate.

F10 — The audit table [LAPTOP]

Run `check_invariance` on each candidate below; record CV under permutation, under scaling, under combined; classify.
Expected invariant: persistence diagrams (β_0, β_1, total persistence) of activation clouds (permutation); weight products; invariant polynomials of weight tensors (traces of products, determinants of Gram matrices); loop products along closed paths.
Expected covariant: singular values of layer maps under scaling; persistence under scaling (measure — do not assume).
Expected artifact: per-unit activations, per-unit gradient×input attributions, top-k unit sets by importance, raw weight norms, single-direction "features" fitted per representative (before alignment).
With alignment on: single-direction features after Procrustes — report whether they become invariant.
Output. `results/audit_table.csv`, `fig/F10_audit_heatmap.png`.
Kill. Any quantity whose classification differs between permutation-only and combined samples is reported with both.

F11 — Real-model audit [5070]

GPT-2-small. Renamings: head permutations, within-head rotations, MLP unit permutations, scalings through GELU where homogeneous (note: GELU is not homogeneous — scalings apply only to LayerNorm-compatible directions; document exactly which scalings are exact and which are approximate). Run F10's candidate list on layer 6 activations over 2000 tokens.
Kill. Any classification that flips between MLP-toy and GPT-2 is flagged, not smoothed.

F12 — Frontier audit [HPCC]

Llama-3-8B or Gemma-2-9B, same procedure, 3 layers. Checkpointed job.

F13 — Ship the auditor + novelty log [LAPTOP]

Public API, docstrings, tests. `NOVELTY.md`: log the searches ("parameter symmetry explanation identifiability", "implementation invariance mechanistic interpretability audit") with dates and top hits.

PAUSE 2 — report F8–F13.

4 Block K — Dark Matter: what the tools miss

Problem attacked: P4 (tool-side), G2. **NOVELTY: OPEN** as application (MP null on transformer *weights* exists, Sept 2025; on SAE *residuals* not found). K7 **NOVELTY: FLAGGED — check before write-up** (Engels et al. may already show residue does not shrink with SAE width — search before claiming).

K1 — Residuals [5070]

GPT-2-small SAEs (SAELens, layers 3/6/9) and Gemma-2-2B Gemma Scope (matching layers). 50k tokens of OpenWebText. $e = x - \mathrm{SAE}(x)$. Save e, x, token ids. Verify reconstruction MSE matches published within 10%.

K2 — Spectral null [5070]

Covariance of e across tokens; eigenvalues; `nulls.marchenko_pastur` with matched d/n and variance $\hat{\sigma}^2$ from the bulk median; count eigenvalues above the Tracy–Widom edge at $p < 0.01$. Structured fraction = sum of outlier eigenvalues / trace.
Pre-registered. >10 outliers per layer.
Kill. Synthetic Gaussian e with matched marginals $\rightarrow \leq 1$ outlier. Also: variance-matched but feature-shuffled real e.

K3 — Topological null [5070]

v1 pipeline on e clouds (PCA-20, 2000 points, 20 bootstraps): total persistence and β_1; ratio to `nulls.feature_shuffle`.
Pre-registered. Ratio > 1.2 with 95% CI excluding 1.

K4 — Predictive content [5070]

Ridge regression $e \rightarrow \Delta$logits (model logits minus SAE-reconstructed-pass logits); held-out R^2 vs shuffled-e R^2.

K5 — Nonlinear residue [5070]

Fit degree-2 polynomial features (random-projection quadratic map, 2000 features) to e; held-out variance explained beyond linear.

K6 — Certified dark matter [5070]

Run K2–K3 statistics through `check_invariance` under model renamings (the SAE dictionary must be transformed consistently). Report invariance class.

K7 — Width scaling [HPCC]

Gemma Scope width ladder (16k $\rightarrow$ 1M latents): K2 structured fraction vs width. **NOVELTY: FLAGGED — check before write-up** — search "Engels dark matter SAE scaling width nonlinear error" first; if occupied, run as replication and say so.

K8 — Frontier residue [HPCC]

Gemma-2-9B Gemma Scope; K2–K5.

K9 — Ship `residual_structure` [LAPTOP]

Unit test: synthetic noise $\rightarrow$ structured fraction < 0.02.

PAUSE 3 — report K1–K9.

5 Block M — Memorization: audit and benchmark the v1 survivor

Status change from earlier roster. Per-concept memorization detection is **NOVELTY: TAKEN — no claim, cite** (noisy-label literature does per-sample detection: MOIT, CrossSplit, ELR; Fisher-rank spectral signature July 2026). M3–M5 run *only* as baseline comparisons; no "first" claim. What remains: audit and benchmark.

M1 — Audit the survivor [5070]

Load v1's battery (21 CNN, 10 LM) and the `memorization_signal` features ($\mathrm{TC}/\mathrm{TC}_{\mathrm{shuffled}}$, persistent entropy). Run each feature through `check_invariance`. Drop any artifact-class component; recalibrate.

M2 — Benchmark [5070]

On the identical battery, compute: susceptibility-to-noisy-labels (Forouzesh et al. 2022), a grokking-without-test complexity proxy, and per-sample noisy-label detectors aggregated to model level (ELR-style loss dynamics). ROC per method; DeLong test for AUC differences.
Verdict rule. If a baseline matches or beats the topological signal, `FINDINGS.md` says so and the library positions the signal as complementary or deprecates it.

M3 — Per-class comparison (baseline study only) [5070]

ResNet-18/CIFAR-10 with 3 of 10 classes label-corrupted. Compute the topological signal per class and ELR/MOIT-style per-sample detection aggregated per class. Report which separates corrupted classes better. Expected: existing methods win. This is a scoping result, not a contribution.

M4 — Cross-family scope [HPCC]

Recalibrate per family (CNN, LM, ViT); report cross-family AUC honestly (v1: 0.00).

6 Block C — Audited Concept Map

Problem attacked: P1. **NOVELTY: FLAGGED — check before write-up** (feature alignment across seeds and RTD/tRSA-style comparison exist; the invariance-*certification* layer on SAE features is likely open — search "SAE feature consistency across seeds" and "gauge fixing sparse autoencoder features" before write-up).

C1 — Concept-method audit [5070]

GPT-2-small layer 6, public SAE. For each renaming sample: recompute SAE features on transformed activations (dictionary transformed consistently), linear probes for 20 standard concepts, and pairwise feature geometry. Through `check_invariance` with `align=False` and `align=True`.

C2 — Certified inventory [5070]

Features whose identity is stable after alignment across all samples form the certified inventory; report fraction certified.

C3 — Certified geometry + Mapper [5070]

On certified features only: invariant relational structure (angles up to gauge, persistence of the feature-vector cloud) and a Mapper graph of activation space with the certified-feature lens. **NOVELTY: FLAGGED — check before write-up** (Mapper on networks exists — TopoAct; cite; contribution is certification).

C4 — Scale [HPCC]

Gemma-2-9B + Gemma Scope.

C5 — Retire v1 P1 claims [LAPTOP]

Write the retirement of FPL↔accuracy and crystallization-order into `AUDIT.md`.

7 Block D — Attribution (anchor only)

NOVELTY: TAKEN — no claim, cite — Implementation Invariance axiom (Sundararajan et al. 2017); axiom satisfaction per method tabulated (survey, July 2026). Path methods pass; LRP/DeepLIFT and neuron-level methods fail.

D1 — Known-answer anchor [5070]

Run integrated gradients (input-level), gradient×input (input-level), and neuron-level attributions through `check_invariance`. Expected: input-level path methods invariant, neuron-level artifact. Purpose: demonstrate the auditor reproduces the known axiom results. No novelty claim; cite.

8 Block B — Bias: certify the real part, retire the rest

Problem attacked: P3. **NOVELTY: FLAGGED — check before write-up** (concept-erasure literature INLP/LEACE is extensive; gauge-invariance of bias directions and subspace-rank-as-erasure-predictor are likely open but adjacent — search before write-up).

B1 — Formal retirement [LAPTOP]

Document v1's precedence-window null (30M/51M/124M, $p \in [0.30, 0.97]$) in `AUDIT.md` with the figures.

B2 — Is the bias direction real geometry? [5070]

GPT-2-small, v1 WinoBias setup. Compute $g_l = \frac{\mu_M - \mu_F}{\|\cdot\|}$ per renaming sample; `check_invariance` with `align=True`. Report direction stability (cosine after alignment) and magnitude covariance.
Kill. Random-split "bias direction" (random halves of the probe set) must be unstable.

B3 — Power curve [5070]

BiasTop permutation test power vs $n \in \{25, 50, 100, 200, 400, 800\}$ per group, with planted divergence of known size. Report n for power 0.8.

B4 — Separability rank [5070]

For v1's gender, COMPAS, toxicity setups: effective rank (participation ratio) of the between-group covariance in the residual stream. Pre-registered: gender $\approx$ rank 1; COMPAS and toxicity high rank. Correlate with v1's erasure success/failure.

B5 — Per-family calibration [HPCC]

BiasTop calibrated on GPT-2, Pythia-410M, Gemma-2-2B with nulls and stated n requirements.

9 Block V — Legibility: what makes a network explainable

Problem attacked: P7, G3. Phase diagram **NOVELTY: OPEN** (Golechha et al. qualitative only). V5 drift **NOVELTY: FLAGGED — check before write-up** (clusterability during training tracked by Filan et al.; the discovery-cost framing is likely new).

V1 — Ensemble [5070]/[HPCC]

$\geq$60 models: 2–3 layer MLPs (widths 64–512) and 2–4 layer transformers (d=128) on modular addition ($p = 113$), sorting (length 8), copying; regularization grid L1 $\in \{0, 10^{-4}, 10^{-3}\}$ $\times$ weight decay $\in \{0, 0.1, 1\}$ $\times$ dropout $\in \{0, 0.2\}$; 10 clusterability-loss models (Golechha recipe); 6 random-label; 6 untrained. Save all checkpoints every 5% of training (for V5).

V2 — Structure [5070]

S4 graphs at thresholds $\{0.01, 0.05, 0.1\}$; Q, treewidth bound, sparsity; degree-preserving rewired null for Q.

V3 — Discovery cost [5070]

S4 frozen pipeline; record time, size, faithfulness, 3-seed stability.

V4 — Phase diagram [LAPTOP]

Success (faithfulness ≥ 0.9 reached) and cost vs $(Q, \text{tw}, \text{sparsity})$; fit a logistic with a changepoint; pre-registered: changepoint model beats linear by BIC > 10.
Figures. `fig/V4_phase_Q.png`, `fig/V4_phase_tw.png`, 2-D (Q, tw) heatmap.

V5 — Drift [5070]

V2 on all saved checkpoints; Q and discovery cost vs training fraction; random-label trajectory as null. Report direction and where the knee (if any) occurs relative to the loss curve.

V6 — v1's loss-invisible reorganization, certified [5070]

Recompute SV peaks on V1 checkpoints; `check_invariance` on SV; feature-shuffle null on the peak. Survives or is retired.

V7 — Pythia drift [HPCC]

Pythia-410M and 1B public checkpoints (all 154): V2 measures on layer-mid attribution graphs from a fixed 2000-token probe. Anyone can verify.

V8 — Ship `legibility_score` [LAPTOP]

Calibrated on V1 with 20% held out; report held-out AUC for "discovery will succeed."

PAUSE 4 — report M, C, D, B, V.

10 Block Z — Topology Beyond Persistent Homology

These deepen the topological foundation. Each carries its own flag.

Z1 — Zigzag persistence across training [5070]

On V1 checkpoints: zigzag module over the sequence of activation clouds (union-of-consecutive construction). Track individual H_1 features' birth/death *in training time*. Compare to v1's per-epoch diagrams: does zigzag identify features the per-epoch film mislabels as births/deaths? **NOVELTY: FLAGGED — check before write-up** (zigzag for time-varying data exists; application to training dynamics — search "zigzag persistence neural network training").

Z2 — Multiparameter persistence [5070]

Scale $\times$ training-fraction bifiltration on selected V1 models; rank-invariant / Hilbert function summaries; test whether the 2-parameter summary predicts V3 discovery cost better than 1-parameter total persistence. **NOVELTY: FLAGGED — check before write-up**.

Z3 — Graph persistent homology of attribution graphs [5070]

Clique-complex PH on S4 graphs (edge-weight filtration); β_1 (cycle structure) as an additional legibility feature in V4. **NOVELTY: FLAGGED — check before write-up** (graph PH is standard; relating it to discovery cost is likely new).

Z4 — Quotient-space connectivity beyond H_0 [LAPTOP]

On F specimens: within each orbit, are explanations connected by paths of valid explanations (mode connectivity in explanation space)? Sample interpolations between orbit members in parameter space; check validity along the path. Reports whether orbits are path-connected (π_0 of the valid set vs H_0 of the quotient). **NOVELTY: OPEN** (as far as found).

Z5 — Persistence under scaling: the covariance law [LAPTOP]

From F10: if persistence is covariant (not invariant) under scaling, derive the exact transformation law empirically and propose a normalized (scale-invariant) persistence statistic; verify via `check_invariance`. Ship as `topology.normalized_persistence`.

11 Block L — Library Release Gates

- L1 Every public function has a `VALIDATION.md` entry; otherwise it is private.
- L2 Every structure detector returns < 0.02 structured fraction / CV-verdict `invariant` on pure noise / anchors.
- L3 Every detector recovers `nulls.planted` structure of known size within 10%.
- L4 Every "better" claim in docs links to its baseline row in `benchmarks/`.
- L5 `CLAIMS.md`: claim $\rightarrow$ experiment $\rightarrow$ evidence file $\rightarrow$ null $\rightarrow$ failure mode $\rightarrow$ novelty status.
- L6 End-to-end on GPT-2-small in one session; timing logged.
- L7 End-to-end on Gemma-2-9B on HPCC; timing logged; what doesn't scale is documented.

L8 `whiteboxtopo` pip-installable; CI green on Python 3.10–3.12.

L9 README pitch: "the auditor," never "opened the black box."

L10 `NOVELTY_BOARD.md` finalized: every claim re-searched within 14 days of release.

12 Novelty Board (status at handbook date)

Claim	Known neighbors	Status
Gauge orbits explain explanation multiplicity	Parameter-symmetry survey 2025; Grigsby functional dimension; Méloux 2025	**OPEN**
Mechanistic invariance audit	Implementation Invariance axiom (Sundararajan 2017) — input-level only	**OPEN** as mechanistic extension; cite axiom
Spectral null on SAE residuals	MP on transformer weights (Sept 2025); Engels dark matter	**OPEN** as application
Dark-matter width scaling	Engels et al.	**FLAGGED**
Quantitative legibility phase diagram	Golechha (qualitative); Adolfi (hardness)	**OPEN**
Modularity drift over training	Filan clusterability	**FLAGGED**
Certified SAE concept inventory	seed-consistency studies; RTD/tRSA	**FLAGGED**
Bias-direction invariance; rank predicts erasure	INLP, LEACE	**FLAGGED**
Per-concept memorization detector	MOIT, CrossSplit, ELR, Fisher-rank 2026	**TAKEN** — baseline only
Attribution invariance	Sundararajan 2017; July 2026 survey	**TAKEN** — anchor only
Zigzag / multiparameter on training	time-varying TDA literature	**FLAGGED**
Explanation-space path connectivity	—	**OPEN** (as found)

13 Pause Schedule and Report Format

PAUSE 0 infrastructure tests → **PAUSE 1** F1–F7 (headline decision) → **PAUSE 2** F8–F13 (audit table) → **PAUSE 3** K (dark matter) → **PAUSE 4** M, C, D, B, V → **PAUSE 5** Z and L.

At each pause, the report is exactly: (1) plain-English paragraph of what was found; (2) the pre-registered thresholds and whether each was met; (3) kill-test outcomes; (4) anything that contradicts the plan's framing, stated first not buried; (5) novelty searches run and their top hits; (6) what is now claimable and what is not.

The last version of this project overclaimed, ran the hard tests, and most of it died. Everything above is built so that cannot happen again: pre-register, kill, baseline, re-search. What survives that is real.